

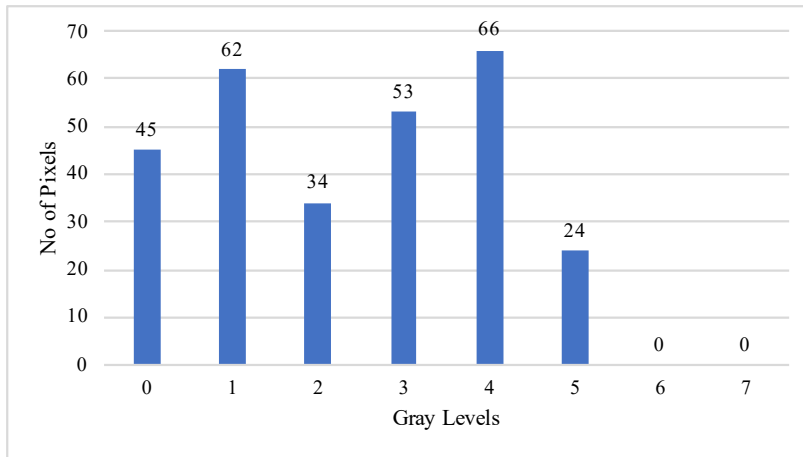
Histogram Equalization Numerical

Q. Perform histogram equalization in given image data.

Image data

r	0	1	2	3	4	5	6	7
n	45	62	34	53	66	24	0	0

Step-01: Plot histogram from given data.



Step-02: Calculate the histogram equalization values.

r	n	PDF	CDF	CDF*(L-1)	Round Off
0	45	0.1584507	0.1584507	1.10915493	1
1	62	0.21830986	0.37676056	2.63732394	3
2	34	0.11971831	0.49647887	3.47535211	3
3	53	0.18661972	0.68309859	4.78169014	5
4	66	0.23239437	0.91549296	6.4084507	6
5	24	0.08450704	1	7	7
6	0	0	1	7	7
7	0	0	1	7	7

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Step-03: Plot output image histogram.

s	0	1	2	3	4	5	6	7
n	0	45	0	96	0	53	66	24

